

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		metallurgists, physicists and chemists are concerned with different properties of heavy metals ✓			1	1		
		(ii)		water becomes contaminated with toxic lead that builds up in body / bioaccumulates			1	1		
		(iii)		lead contamination in road-side soil at all distances is much greater in towns than in the country ✓ lead contamination in road-side soil decreases between 12m and 42m from the centre of the road in the countryside ✓			2	2		
		(iv)		it took 15 years for paint and petrol to become lead free ✓			1	1		
	(b)	(i)		6 PbCO ₃ + O ₂ → 2 Pb ₃ O ₄ + 6 CO ₂		1		1	1	
		(ii)		Pb ₂ O ₃ (3) if answer incorrect award (1) for each of following mass of oxygen = 1.18 g Pb : O ratio is 0.0491 : 0.07375 ecf possible		3		3	3	
				Question 8 total	0	4	5	9	4	0